

Reader Agency and the Recovery of Permutation Structure

Fixed Cuts, Iteration, and Movable Cuts

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Abstract

A permutation T of N cells is observed through a *cut*: the cells are divided into blocks and only the block-to-block counts are reported. A single such reading leaves a large set of permutations indistinguishable. We compare three ways of learning more: reading repeatedly as T iterates, refining the cut, and moving the cut while keeping its coarseness fixed.

For balanced two-block cuts the whole table is determined by one number, the count of cells that stay in the first block. That number depends only on the underlying undirected edges of T , so reversing the orientation of any subset of T 's cycles leaves every balanced two-block reading unchanged, for every cut and every N . The group of such reversals is $(\mathbb{Z}_2)^k$, where k counts the cycles of length at least three, so its orbits have size 2^k .

We measure what each capability recovers. For $N = 8$, a fixed two-block reader watching all fifteen powers of T leaves 10.066 bits hidden and never determines the cycle type. The same reader taking a single reading, but allowed to place its cut in all thirty-five ways, leaves 1.218 bits and does determine the cycle type. Adding iteration on top of the movable cut changes nothing at all: both give exactly 18,155 classes. So agency dominates duration, and once cuts may move, duration is worth nothing.

We prove that for every even $N \geq 8$ the classes of the movable-cut reader are exactly the cycle-reversal orbits, so the reader recovers the unoriented cycle structure of T and loses precisely the orientations. The result fails for $N = 4$ and $N = 6$, and those are the only exceptions. At $N = 8$ the classes are 764 singletons, 15,004 pairs and 2,387 quadruples. We also give the passive causal quotient at several resolutions, where no coarse reader ever determines the cycle type, the number of cycles, or the order of T , however long it watches.

Keywords: permutations, contingency tables, double cosets, causal states, Nerode equivalence, cycle orientation, observability.

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1. Introduction

Let T be a permutation of the cells $0, 1, \dots, N - 1$. A *cut* is a partition of the cells into blocks; the reader associated with a cut reports the table of block-to-block counts, and nothing else. Two permutations with the same table lie in the same double coset of the Young subgroup of the cut, so the reader's resolution is exactly that double coset [3, 5].

A companion paper [4] determines what a single reading of a single cut reveals: an exact information profile, the fiber sizes, and the fact that no cut short of full resolution determines the cycle structure. This paper asks what an observer can do about that. Three capabilities are available, and they are not equivalent.

- **Duration.** Read the same cut repeatedly as T iterates, observing the tables of $T, T^2, T^3, \dots$
- **Resolution.** Use a finer cut, with more blocks.
- **Agency.** Keep the coarseness fixed but move the cut, reading the same permutation through different divisions of the same cells.

We measure all three on the same object. The ranking is unambiguous, and duration comes last.

Main results.

1. **Reversal invisibility (Proposition 2.3, Theorem 3.1).** For balanced two-block cuts, a reading depends only on the undirected edge multiset of T . Reversing any subset of T 's cycles therefore changes no reading, under any cut, for any N . The reversal group is $(\mathbb{Z}_2)^k$ with k the number of cycles of length at least three, and orbits have size 2^k .
2. **The hierarchy (Section 4).** For $N = 8$: a fixed cut leaves 13.657 bits hidden; adding all iterations leaves 10.066; a movable cut alone leaves 1.218; and a movable cut with iteration leaves 1.218 again.
3. **Exact quotient (Theorem 4.3).** For every even $N \geq 8$, two permutations have the same signature if and only if they lie in the same cycle-reversal orbit. The reader recovers the unoriented cycle structure and loses exactly the orientations.
4. **The exceptions (Proposition 4.5).** At $N = 4$ and $N = 6$ the statement fails, and those are the only even cases where it does.
5. **The passive causal quotient (Section 5).** Watching a fixed coarse cut forever recovers between three and four bits regardless of resolution, and never determines the cycle type, the number of cycles, or the order of T .

2. Setup

Write S_N for the permutations of $\{0, \dots, N - 1\}$ and $\lambda(T)$ for the cycle type of T .

Definition 2.1. A *cut* is a map ℓ from cells to blocks. Its reading of T is the table $C_\ell(T)_{ij} = |\{x: \ell(x) = i, \ell(T(x)) = j\}|$. A cut is *balanced two-block* when N is even and each of the two blocks holds $N/2$ cells. Write $\mathcal{B}_N$ for the set of balanced two-block cuts, of which there are $\binom{N}{N/2}/2$.

For a balanced two-block cut, both row sums and both column sums equal $N/2$, so the table is determined by its top-left entry

$$r_\ell(T) = |\{x: \ell(x) = 0, \ell(T(x)) = 0\}|.$$

The *signature* of T is the vector $(r_\ell(T))_{\ell \in \mathcal{B}_N}$.

Definition 2.2 (reversal). Let T have cycles $c_1, \dots, c_r$. For a subset A of those cycles, $\rho_A(T)$ is the permutation obtained by reversing the orientation of each cycle in A and leaving the others alone. The *reversal orbit* of T is $\{\rho_A(T)\}$ over all A .

Proposition 2.3. For every balanced two-block cut ℓ and every subset A ,

$$r_\ell(\rho_A(T)) = r_\ell(T).$$

Proof. View T as the directed graph with an arc $x \rightarrow T(x)$ for every cell, and let $E(T)$ be the multiset of unordered pairs $\{x, T(x)\}$. Reversing a cycle replaces its arcs $x \rightarrow T(x)$ by $T(x) \rightarrow x$, so it preserves $E(T)$. Now $r_\ell(T)$ counts the arcs whose tail and head both lie in block 0, which is the number of pairs of $E(T)$ with both endpoints in block 0, since an arc lies inside block 0 exactly when the corresponding unordered pair does. Hence r_ℓ depends only on $E(T)$. ◻

The restriction to *balanced* two-block cuts matters. For an unbalanced cut the two off-diagonal entries need not be equal, and reversal can exchange them.

3. The reversal group

Theorem 3.1. The maps ρ_A form a group isomorphic to $(\mathbb{Z}_2)^k$, where k is the number of cycles of T of length at least three. The reversal orbit of T has exactly 2^k elements.

Proof. Reversal acts independently on each cycle and is an involution on each, so the maps ρ_A commute and each squares to the identity. A cycle equals its own reverse precisely when it has length one or two, so only the cycles of length at least three contribute a nontrivial factor, giving $(\mathbb{Z}_2)^k$. The action is free on those k coordinates, so the orbit has 2^k elements. ◻

Reversal preserves cycle lengths, so it acts within a conjugacy class. Global inversion $T \mapsto T^{-1}$ is the single element ρ_A with A all cycles; it is one element of a group of order 2^k , not the whole group. This distinction matters: a reader blind to reversal loses k bits, not one.

The orbit-size distributions were checked against 2^k for every permutation with $N \leq 8$, with no exceptions.

4. The hierarchy

Fix N even. We compare four observers, all restricted to balanced two-block readings.

- O_1 , **fixed cut**: one reading through one cut.
- O_2 , **duration**: readings of $T, T^2, \dots, T^H$ through one fixed cut, with H at least the largest order of a permutation of N cells.
- O_3 , **agency**: one reading through each cut in $\mathcal{B}_N$.
- O_4 , **both**: readings of all powers through all cuts.

Each observer induces an equivalence on S_N . Write O for the observation. For the uniform distribution on S_N , the hidden information of an observer is the conditional entropy $H(T \mid O)$, equivalently the expected value of $\log_2$ of the size of the observation class containing T .

Table 1. What each capability recovers. $N = 6$ uses 10 cuts and horizon 6; $N = 8$ uses 35 cuts and horizon 15.

observer	$N = 6$ classes	$N = 6$ bits hidden	$N = 8$ classes	$N = 8$ bits hidden	cycle type
O_1 fixed cut	4	8.023	5	13.657	no
O_2 iterates, fixed cut	14	5.770	40	10.066	no
O_3 every cut	378	1.006	18,155	1.218	yes
O_4 iterates, every cut	378	1.006	18,155	1.218	yes

Three features stand out. Moving the cut recovers far more than iterating: at $N = 8$, agency leaves 1.218 bits where duration leaves 10.066. Agency determines the cycle type and duration does not. And $O_4 = O_3$ exactly, so once the cut can move, iteration contributes nothing.

The last point has a structural explanation rather than a numerical one, and for two distinct reasons rather than one. Every balanced two-block reading of every power is invariant under reversal, because Proposition 2.3 applies to T^t , and because reversing the cycle orientations of T replaces T^t by its inverse on the cycles involved, which leaves the undirected edge multiset of T^t unchanged. An observation that is invariant under a group cannot separate that group's orbits, no matter how many times it is repeated. That is the whole story for $N \geq 8$, where Theorem 4.3 makes O_3 's classes exactly the reversal orbits. At the two exceptional N it is not the whole story, since some of O_3 's classes there are strictly larger than a reversal orbit (Proposition 4.5); §4.1.1 gives the second, independent reason duration cannot split those either.

4.1 The exact quotient

Write $B(T)$ for the symmetric matrix of the undirected edge multiset: for $u \neq v$, B_{uv} is the number of times the pair $\{u, v\}$ occurs among the pairs $\{x, T(x)\}$, and $B_{uu} = 1$ if u is a fixed point and 0 otherwise. Entries satisfy $B_{uv} \in \{0, 1, 2\}$ and $B_{uu} \in \{0, 1\}$, and every vertex has degree two: $\sum_{v \neq u} B_{uv} + 2B_{uu} = 2$.

Lemma 4.1. $r_\ell(T) = \sum_{u < v \in A} B_{uv} + \sum_{u \in A} B_{uu}$, where A is block 0 of ℓ . Two permutations have the same signature if $B(T) = B(T')$, and $B(T) = B(T')$ if and only if T and T' lie in the same reversal orbit.

Proof. The first statement restates Proposition 2.3: an arc lies inside A exactly when its unordered pair does. For the second, a permutation is recovered from B by orienting each component: a loop gives a fixed point, a doubled edge forces a transposition, and an undirected cycle of length at least three admits exactly two orientations. The permutations sharing an edge matrix are therefore exactly the reversal orbit. ▫

Lemma 4.2 (swap identity). Let $D = B(T) - B(T')$ and suppose T, T' have the same signature. Then there are numbers $a_0, \dots, a_{N-1}$ with

$$D_{uv} = a_u + a_v \ (u \neq v), \quad D_{uu} = -\frac{N-2}{2}a_u, \quad \sum_u a_u = 0.$$

Proof. Let A be balanced, $a \in A$, $b \notin A$, and let A' be A with a replaced by b . Both are balanced, so $e_A(D) = e_{A'}(D) = 0$, and subtracting gives

$$\sum_{v \in S} (D_{av} - D_{bv}) = D_{bb} - D_{aa},$$

where S is A with a removed. Here S ranges over all $(N/2 - 1)$ -element subsets of the cells other than a and b , and $1 \leq N/2 - 1 \leq N - 3$ for $N \geq 4$, so exchanging one element of S shows that $D_{av} - D_{bv}$ is the same for every $v \notin \{a, b\}$; call it c_{ab} .

Fix the reference index 0 and put $c_u = c_{u0}$ and $h_v = D_{0v}$. For $u, v \notin \{0\}$ distinct, $D_{uv} = h_v + c_{uv}$ and symmetry gives $h_v - c_v = h_u - c_u$, a constant k . Hence $D_{uv} = c_u + c_v + k$ for all $u \neq v$, with $c_0 = 0$. Substituting into the degree identity $\sum_{v \neq u} D_{uv} + 2D_{uu} = 0$, which holds because T and T' both have all degrees two, gives $2D_{uu} = -[(N-2)c_u + C + (N-1)k]$ with $C = \sum_v c_v$. Writing $a_u = c_u + k/2$ the terms in k cancel and these become $D_{uv} = a_u + a_v$ and $2D_{uu} = -[(N-2)a_u + S]$ with $S = \sum_u a_u$.

It remains to show $S = 0$, and the degree identity cannot supply it: substituting the two displayed forms back into $\sum_{v \neq u} D_{uv} + 2D_{uu}$ gives $(N-2)a_u + S - [(N-2)a_u + S] = 0$ for any S . So we use one balanced equation. Take any balanced A and write $n = N/2$ and $S_A = \sum_{u \in A} a_u$. Then

$$e_A(D) = \sum_{u < v \in A} (a_u + a_v) + \sum_{u \in A} D_{uu} = (n-1)S_A - \frac{1}{2}[(N-2)S_A + nS] = -\frac{n}{2}S,$$

since $(n-1)S_A = \frac{N-2}{2}S_A$. As $e_A(D) = 0$ and $n > 0$ we get $S = 0$, and then $D_{uu} = -\frac{N-2}{2}a_u$. ▫

Theorem 4.3. Let $N \geq 8$ be even. Two permutations of $[N]$ have the same signature if and only if they lie in the same cycle-reversal orbit.

Proof. Sufficiency is Proposition 2.3. For necessity, take D and a as in Lemma 4.2 and put $q = \frac{2}{N-2}$.

Since $B_{uu}, B'_{uu} \in \{0,1\}$ we have $D_{uu} \in \{-1,0,1\}$, so $\frac{N-2}{2}a_u \in \{-1,0,1\}$ and therefore $a_u \in \{-q, 0, q\}$ for every u . Since B and B' are integer matrices, $D_{uv} = a_u + a_v$ is an integer for all $u \neq v$.

For $N \geq 8$ we have $0 < q \leq \frac{1}{3}$, so $q \notin \mathbb{Z}$ and $2q \notin \mathbb{Z}$ unless $N-2$ divides 4, which fails for $N \geq 8$. Suppose some $a_u \neq 0$. If some $a_v = 0$ then $D_{uv} = \pm q \notin \mathbb{Z}$, a contradiction. So every $a_u = \pm q$; since $\sum_u a_u = 0$, both

signs occur $N/2 \geq 4$ times, and two entries of the same sign give $D_{uv} = \pm 2q \notin \mathbb{Z}$, again a contradiction. Hence $a \equiv 0$, so $D = 0$ and $B(T) = B(T')$. Lemma 4.1 finishes the proof. ▫

Corollary 4.4. For even $N \geq 8$, the family of all balanced two-block cuts determines the unoriented cycle structure of TT exactly and determines nothing about the orientations. The classes have size $2k2^k$, where k is the number of cycles of TT of length at least three, and the number of classes is the number of labeled 2-regular undirected multigraphs on $[N]$, with loops and doubled edges allowed, arising as the edge matrices $B(T)B(T)$.

Proposition 4.5 (the exceptions). For $N = 4$ and $N = 6$ the conclusion of Theorem 4.3 fails, and these are the only even exceptions.

Proof. The argument above breaks exactly when q or $2q$ is an integer, that is when $N - 2$ divides 4: $N = 4$ gives $q = 1$ and $N = 6$ gives $q = \frac{1}{2}$ with $2q = 1$. Both admit nonzero solutions, and exhaustive computation exhibits them. At $N = 4$ there are 11 classes, four larger than a reversal orbit, including one class of size 8 containing all eight 3-cycles; the difference has $a = (-1, -1, 1, 1)$ up to relabelling. At $N = 6$ there are 378 classes, ten larger than a reversal orbit, all of cycle type $(3, 1, 1, 1)$, with a equal to $\pm \frac{1}{2}$ in three coordinates each. ▫

At $N = 8$ the 18,155 classes consist of 764 of size 1, 15,004 of size 2 and 2,387 of size 4, matching 2^k for $k = 0, 1, 2$. The singletons are the involutions of S_8 . The classes of size 4 are those with exactly two cycles of length at least three:

Table 2. The size-4 classes at $N = 8$.

cycle type	classes of size 4
(5,3)	672
(4,4)	315
(4,3,1)	840
(3,3,2)	280
(3,3,1,1)	280
total	2,387

The cut family grows quickly, with $|\mathcal{B}_N| = 3,10,35,126$ for $N = 4, 6, 8, 10$, but by Theorem 4.3 the obstruction at $N = 4$ and $N = 6$ is arithmetic rather than a shortage of cuts. At $N = 10$ we confirmed the theorem computationally on all 240 permutations of cycle type $(3, 1^7)$, the family that fails at small N , and on 60,000 uniformly random permutations.

4.1.1 Why exactly these two

Note added in review. The proof of Proposition 4.5 above stops at exhibiting the exceptional solutions computationally. The remark and corollary below give the exact mechanism, so that the scope of the exceptions and the survival of the mergers under duration (used in the hierarchy discussion two subsections earlier) are both established rather than only observed.

Remark 4.5.1 (structural account of the exceptions). Write $m = \frac{N}{2}$. Every exceptional merger found above pairs an m -cycle on some m -subset S with the m -cycle on the complementary set F , each permutation fixing the other set pointwise. Call these T and T' .

Because T moves only S and T' moves only F , both $B(T)$ and $B(T')$ are block-diagonal in the S/F split: on S , $B(T)$ is the simple graph of the m -cycle and $B(T')$ is the identity; on F the roles reverse; neither matrix has an S – F entry. Lemma 4.2 needs $D_{uv} = a_u + a_v$ to take the same value for every pair inside S — so the m -cycle graph on S must be complete, since one missing edge would force a zero where the identity needs the same nonzero constant that every other pair gets. A cycle graph on m vertices is complete exactly at $m = 2$ (a single doubled edge) and $m = 3$ (a triangle); at $m \geq 4$ it always omits a chord. So the mechanism reaches exactly $N = 4$ and $N = 6$ and no further — this is the exact reach of the identity, not a search radius.

Checked against the census in the proof of Proposition 4.5: every one of the ten anomalous classes at $N = 6$ is exactly this pairing, for all ten complementary triples, and the three transposition mergers at $N = 4$ are the same pairing at $m = 2$. The remaining $N = 4$ anomaly — the one class of size 8 holding all eight 3-cycles — has support size 3 against blocks of size 2, not support size m against blocks of size m , so it is not an instance of this identity. It stands as before, confirmed only by direct enumeration.

Corollary 4.5.2 (the mergers survive duration). The identity above is preserved under iteration, in two regimes. Since $m \in \{2, 3\}$ is prime, T^t is again an m -cycle on S , and T'^t an m -cycle on F , exactly when $m \nmid t$; the block-diagonal computation then returns the same difference D as at $t = 1$. When $m \mid t$ both are the identity and $D = 0$. Either way the two permutations have the same signature at every t , so the merger of Proposition 4.5 is fixed by the identity that produces it, rather than merely left unbroken by whatever horizon was tried.

Verified at $N = 4$ and $N = 6$ over all t through $4m$: the signatures agree at every t ; D takes exactly one nonzero value across all t with $m \nmid t$; and $D = 0$ at every t with $m \mid t$. *Correction.* An earlier draft of this corollary asserted that D is the identical nonzero matrix at every t . That is false on the multiples of m , where both permutations collapse to the identity; the conclusion is unchanged because the readings still agree there, but for the second reason rather than the first.

4.2 How many cuts are needed

Theorem 4.3 uses the whole family $\mathcal{B}_N$, whose size grows exponentially. Far fewer cuts suffice.

Proposition 4.6 (counting bound). A family of f cuts assigns to each permutation a vector of f readings, each in $\{0, 1, \dots, N/2\}$, so it separates at most $(N/2 + 1)^f$ classes. Separating all of them therefore requires at least $\lceil \log_{N/2+1} K \rceil$ cuts, where K is the number of classes: 3, 5 and 7 for $N = 4, 6, 8$.

Searching for small sufficient families gives upper bounds of 3, 9 and 18 for $N = 4, 6, 8$, by randomized selection followed by pruning; a greedy rule gives 3, 9 and 19. So at $N = 8$ between 7 and 18 of the 35 cuts are needed, and the exact minimum is not known (Problem 2).

The proof of Lemma 4.2 uses only pairs of cuts differing by a single swap, which suggests the family consisting of one base cut together with all $(N/2)^2$ cuts obtained from it by one swap. At $N = 8$ that family has 17 cuts and yields only 14,457 classes, short of 18,155, so it does not suffice. The swap relations are

enough to derive the form of D only when they are available for every pair of cells against every background, which that family does not provide.

5. The passive causal quotient

Duration is weak but not empty. We measure it separately across resolutions. Let C_m be the reader with m equal contiguous blocks. Define two equivalences: $T \sim_m T'$ when $C_m(T) = C_m(T')$, and $T \approx_m T'$ when $C_m(T^t) = C_m(T'^t)$ for all $t \geq 1$. The second is the passive causal quotient in the sense of Nerode equivalence and causal states [1, 2]: two permutations are identified exactly when no future passive reading can separate them.

Table 3. Watching a fixed cut forever. Horizons are 6 for $N = 6$ and 15 for $N = 8$, each at least the largest order of a permutation.

N	m	fibers	causal classes	bits hidden now	bits hidden forever	bits recovered	cycle type
6	1	1	1	9.492	9.492	0.000	no
6	2	4	14	8.023	5.770	2.253	no
6	3	21	87	5.400	3.067	2.333	no
8	1	1	1	15.299	15.299	0.000	no
8	2	5	40	13.657	10.066	3.591	no
8	4	282	2,457	7.429	4.052	3.376	no

Two observations. The amount recovered is nearly independent of resolution: 3.591 bits at $m = 2$ against 3.376 at $m = 4$, although the instantaneous fibers differ by a factor of 56. And the recovery is bounded well away from the hidden total at every resolution, so most of what a coarse cut hides stays hidden forever.

Proposition 5.1. For $N \leq 8$ and every proper divisor m of N , the passive causal quotient determines neither the cycle type, nor the number of cycles, nor the order of T .

Proof. Exhaustive computation over all permutations, with a horizon at least the largest order, for every one of the twelve pairs (N, m) with $N \leq 8$. Table 3 tabulates the two largest N ; the remainder are in the notebook. ▀

This strengthens the single-reading statement of [4, Theorem 6.1] from one observation to all observations. The failure to determine even the order is worth noting, since the sequence of tables is periodic with period dividing the order; the period of the table sequence can be a proper divisor of the order, and different orders can produce identical sequences.

6. Discussion

The three capabilities are not interchangeable, and their ranking is the opposite of the usual intuition that watching longer is the cheap way to learn more. On this object, moving the instrument recovers what unlimited duration cannot, and after moving the instrument, duration adds precisely zero.

The reason is that duration repeats an observation that is already invariant under a group. Reversal invisibility is not a shortage of samples; it is a symmetry of the observable. An observer facing such a symmetry has only one remedy, namely a different observable, and changing the cut supplies one while iterating does not.

The residual is also informative. What survives every balanced two-block cut at $N = 8$ is exactly the orientation of the cycles, which is k independent bits rather than the single bit of global inversion. Direction, in this setting, is local to each loop.

7. Open problems

1. Give a closed form, or asymptotics, for the number of classes of the movable-cut reader, which by Corollary 4.4 counts the undirected 2-regular multigraphs realized by permutations. The values are 11, 378 and 18,155 for $N = 4, 6, 8$.
2. Determine the smallest sub-family of $\mathcal{B}_N$ that already forces Theorem 4.3. The proof uses only swaps of a single element between blocks, so far fewer than all $\binom{N}{N/2}/2$ cuts should suffice.
3. Extend Table 3 to m -block movable cuts for $m > 2$, and determine at which m the residual orientation ambiguity disappears.
4. Determine the asymptotics of the bits recovered by duration alone. The measured values are near 3.4 bits at $N = 8$ and near 2.3 at $N = 6$, with little dependence on resolution.
5. For unbalanced two-block cuts, the two off-diagonal entries differ and Proposition 2.3 no longer applies. Measure what such a reader recovers.

References

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Appendix. Computation

All computations use exact integer arithmetic. A self-contained notebook, using only the Python standard library, reproduces every number in this paper and runs in about a minute in its default setting.

Statement	Check	Range
Proposition 2.3	signatures under all cycle reversals	$N = 6,8,10$, sampled
Theorem 3.1	orbit size against 2^k	all permutations, $N \leq 8$
Table 1	four observers, class counts and hidden bits	all permutations, $N = 6,8$
Lemma 4.2, Theorem 4.3	the form of D , and the integrality obstruction	$N = 4,6$ examples; N even up to 20
Corollary 4.4, Table 2	classes against reversal orbits	all 40,320 permutations, $N = 8$
Proposition 4.5	extra collisions and their α -vectors	all permutations, $N = 4,6$
Proposition 5.1, Table 3	passive causal quotient	all permutations, every $N \leq 8$ and proper divisor m (12 pairs)
Theorem 4.3 check	signature collisions	$N = 10$: all of type $(3, 1^7)$, and 60,000 random
Remark 4.5.1	every anomaly is a complementary-support pair	all anomalous classes, $N = 4,6$
Corollary 4.5.2	signatures and D at every iterate	$N = 4,6$, all t through $4m$
Proposition 4.6	counting bound, search, swap family	$N = 4,6,8$